FACTOR TREE ANALYSIS FOR OCCURRENCE - Why the problem occurred ?

PROBLEM: 产品外观 - 焊接件 / 焊接件变形/断裂 - 端面变形 - 不稳定

4M [2]	FACTORS [3]	STANDARD [5]		Factor VALUE on OK PARTS [5] (Good Factor / Accepted factor as reference / NO)	Factor VALUE on BAD PARTS [7] (Bad Factor / Unacceptable factor as reference / NO)	JUDGMENT: Bad now	Bad past	Blind spot	Potential Root Causes	Ishikawa WHY until you arrive at the real root cause: 5 WHYs	Countermeasure?
MACHIN	电阻焊端面修磨 Face dressing of the weld cap for circular resistance	Has you the standard value and tolerance?	角度: 0.5-5.0Deg Angle: 0.5-5.0Deg					N/A		1st WHY	
		Where was the standard value defined (type, number and index of the document) ?	文件: 电阻焊设计规范手册 (文件编号: WI0100001) Document: Circular Welding Operation Standard Book, Document Number: WI0100001	以10万次生产计算是由动模来定位, 定位气压, 4.5倍的最大值 Based on 100k times, the equipment inspection of each production process was done and there were no abnormalities. The actual air pressure was 4.5 MPa with evidence attached	以10万次生产计算是由动模来定位, 定位气压, 4倍的最大值 Based on 100k times, the equipment inspection of each production process was done and there were no abnormalities. The actual air pressure was 4 MPa with evidence attached					2nd WHY	
		Has you the standard value measured or verified?	非校准表单有效期与实际值, 必须保证在有效期内 Yes, the air pressure, compressed air, and the machine are valid							3rd WHY	
		Has you the frequency of measurement or verification?	1次/Where per item							4th WHY	
										5th WHY	
MACHIN	表面粗糙度检查 The smoothness of the welding metallic - break box	Has you the standard value and tolerance?	最佳范围: 2.0-5.0Deg 容差范围: 0.0-10.0Deg Best Range value: 2.0-5.0Deg Acceptable Range value: 0.0-10.0Deg	对表面粗糙度进行目测检查, 检查焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box surface roughness was done and there was no compliance with the gauge and no compliance with the step gauge	对表面粗糙度进行目测检查, 检查焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box surface roughness was done and there was no compliance with the gauge and no compliance with the step gauge				N/A	1st WHY	
		Where was the standard value defined (type, number and index of the document) ?	文件编号: 检验规范手册 (文件编号: WI-0100-003) Document: Inspection Instruction Manual (alloucation Number: WI-0100-003)							2nd WHY	
		Has you the standard value measured or verified?	使用前/每周/每月/每季度, 必须/每年 Yes a certain gauge for measuring, 3D/2D gauge							3rd WHY	
		Has you the frequency of measurement or verification?	每批次抽样, 每周3-5次抽检 Use each batch of incoming materials, random inspection is conducted on Wednesday and on Friday							4th WHY	
										5th WHY	
MACHIN	螺纹环端面垂直度 Check the dimensions of the incoming material - break box	Has you the standard value and tolerance?	最佳范围: 2.0-5.0Deg 容差范围: 0.0-10.0Deg Break Box width: 2.0-5.0Deg Acc breaking height: 0.0-10.0Deg	对螺纹环端面垂直度进行测量, 检查焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box surface roughness was done and there was no compliance with the gauge and no compliance with the step gauge	对螺纹环端面垂直度进行测量, 检查焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box surface roughness was done and there was no compliance with the gauge and no compliance with the step gauge				N/A	1st WHY	
		Where was the standard value defined (type, number and index of the document) ?	文件: 检验规范手册 Process inspection instruction manual and inspection record							2nd WHY	
		Has you the standard value measured or verified?	使用前/每周/每月/每季度 Yes, the air pressure, compressed air, and the machine are valid							3rd WHY	
		Has you the frequency of measurement or verification?	每批次/Each lot							4th WHY	
										5th WHY	
MACHIN	螺纹环端面垂直度 Check the dimensions of the welded product	Has you the standard value and tolerance?	最佳范围: 2.0-5.0Deg 容差范围: 0.0-10.0Deg Break Box width: 2.0-5.0Deg Acc breaking height: 0.0-10.0Deg	对螺纹环端面垂直度进行测量, 检查焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box surface roughness was done and there was no compliance with the gauge and no compliance with the step gauge	对螺纹环端面垂直度进行测量, 检查焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box surface roughness was done and there was no compliance with the gauge and no compliance with the step gauge				N/A	1st WHY	
		Where was the standard value defined (type, number and index of the document) ?	过程检验指导书编制记录 Process inspection instruction manual and inspection record							2nd WHY	
		Has you the standard value measured or verified?	使用前/每周/每月/每季度 Yes, the air pressure, compressed air, and the machine are valid							3rd WHY	
		Has you the frequency of measurement or verification?	每批次/Each lot							4th WHY	
										5th WHY	
MACHIN	焊接后清理 Post-welding cleaning	Has you the standard value and tolerance?	不得有焊渣/毛刺/飞溅/凸起/凹陷/变形/表面粗糙度 Is the product to be welded should be kept clean, free of dirt, and free from any foreign matter. The product to be welded should be kept in the correct position, distance from the welding machine, etc.	焊接件表面清理目测, 检查焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box was done and there was no compliance with the step gauge	焊接件表面清理目测, 检查焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box was done and there was no compliance with the step gauge				焊接过程中, 为防产生氧化物和飞溅物, 在焊接过程中, 焊接件表面会涂抹防飞溅剂 During the welding process, the product to be welded should be kept clean, free of dirt, and free from any foreign matter. The product to be welded should be kept in the correct position, distance from the welding machine, etc. Anti-fume spray is used when welding to prevent oxidation and spatter.	1st WHY	焊接产品的表面不干净, 有焊渣, 毛刺, 飞溅物, 凸起, 凹陷, 变形, 表面粗糙度不合格 The non-qualified surface finish of the welding product is unacceptable and contains slag, burrs, spatter, bulges, depressions, deformation, etc.	必须保证焊接件表面干净, 无焊渣, 毛刺, 飞溅物, 凸起, 凹陷, 变形, 表面粗糙度不合格, 焊接件应放置在正确位置, 距离焊接机等位置, 焊接过程中, 为防产生氧化物和飞溅物, 在焊接过程中, 焊接件表面会涂抹防飞溅剂 The product to be welded is close to the welding fixture, and Office are produced under normal operations. The appearance is fair, with no burrs, spatter, bulges, depressions, or deformation on the break box plate. There are no foreign objects or scale residues on the outside and inside of the break box.
		Where was the standard value defined (type, number and index of the document) ?	过程检验指导书编制记录 Process inspection instruction manual and inspection record							2nd WHY	
		Has you the standard value measured or verified?	目测/visual							3rd WHY	
		Has you the frequency of measurement or verification?	100k/100k/Batch lot/1k/1k 100k self-inspection / first and last piece / 1 piece every 2 hours							4th WHY	
										5th WHY	
MACHIN	焊接头角度 Checking of welding stop tip	Has you the standard value and tolerance?	焊接头角度参数表/Welding Tip The welder 100k welding stop tip 100000 times	焊接件检验, 焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box was done and there was no compliance with the step gauge	焊接件检验, 焊接件表面是否有毛刺, 焊渣等缺陷 By Batch 100k, the process inspection record shows that the measurement of the break box was done and there was no compliance with the step gauge				N/A	1st WHY	
		Where was the standard value defined (type, number and index of the document) ?	焊接参数表格/Welding Parameter Sheet WI0100001 Manual Soldering Operation Manual							2nd WHY	
		Has you the standard value measured or verified?	设备点检表 Equipment Maintenance Schedule							3rd WHY	
		Has you the frequency of measurement or verification?	每周/Per week							4th WHY	
										5th WHY	
MACHIN	焊机头的清洁 Cleaning of welding stop tip	Has you the standard value and tolerance?	焊接机设备保养手册及焊机头清洁维护保养记录 There is an requirement for maintenance and cleaning of the welder of the welding stop tip in the welding operation manual	焊接工人检查焊机, 焊机的Stop, 焊机 飞溅, 焊渣, 凸起, 凹陷, 变形, 表面粗糙度 Welding Operation Instruction Manual, Welding Stop, welding Office, every welding worker has	焊接工人检查焊机, 焊机的Stop, 焊机 飞溅, 焊渣, 凸起, 凹陷, 变形, 表面粗糙度 Welding Operation Instruction Manual, Welding Stop, welding Office, every welding worker has				N/A	1st WHY	检查焊接飞溅 Basic explosion welding spatter
		Where was the standard value defined (type, number and index of the document) ?	WI0100001焊工操作设备操作手 WI0100001 Manual Soldering Operation Manual							2nd WHY	焊接头角度的不清洁 The welder of the welding stop tip has not been cleaned
		Has you the standard value measured or verified?	目测 Visual							3rd WHY	焊接机设置等参数未检查/焊接头清洁维护保养记录 Welding equipment parameters not checked in welding stop tip in the welding operation manual
		Has you the frequency of measurement or verification?	每周 Per week							4th WHY	
										5th WHY	
MACHIN	螺母焊接 Nutter projection	Has you the standard value and tolerance?	焊接件表面应有光洁度	焊接件检验, 焊接件表面是否有毛刺, 焊渣等缺陷, 焊点是否有虚焊, 焊点是否有焊接不牢固现象 The appearance is good, with no short-circuit, no burrs, no welding slag, no implosion causes, no gaps, and the number of welded joints is correct. There is no fine crusher or oxidation inside the break box.	焊接件检验, 焊接件表面是否有毛刺, 焊渣等缺陷, 焊点是否有虚焊, 焊点是否有焊接不牢固现象 During the welding process, resin inside the welding machine, welding Office, and welding workers, the welding product to the fixture, resin will splash into the break box				N/A	1st WHY	
		Where was the standard value defined (type, number and index of the document) ?	WI0100001焊工操作设备操作手 WI0100001 Manual Soldering Operation Manual							2nd WHY	
		Has you the standard value measured or verified?	目测 Visual							3rd WHY	
		Has you the frequency of measurement or verification?	每周 Per week							4th WHY	
										5th WHY	
MAN	人员资质 Personnel qualification	Has you the standard value and tolerance?	以1级操作员为准, 在1级管理之下的3级 2 welding operation with a skill level of 4 years	通过各级操作员考核管理体系, 对焊接工艺, 焊接工艺参数, 焊接工艺操作指导书, 焊接工艺检验指导书, 焊接工艺检验记录, 焊接工艺操作记录, 焊接工艺检验记录, 焊接工艺操作记录, 焊接工艺检验记录, 焊接工艺操作记录